

12 Vectors and Geometry of Space

12.1 3D Coordinate Systems

Prerequisite information for Section 12.1:

1. Give an equation for the circle with a diameter between the points, $(4, 9)$ and $(7, -3)$. Please show your work for finding the center and radius of this circle. **Note: This is a TWO-dimensional problem**—just like you did in pre-calculus!

$$\boxed{\frac{153}{4} = (x - 5.5)^2 + (y - 3)^2}$$

$$\begin{aligned} \sqrt{(7-4)^2 + (-3-9)^2} &= \sqrt{3^2 + 12^2} = \sqrt{9 + 144} \\ &= \sqrt{153} \\ \Rightarrow r^2 &= \frac{153}{4} \end{aligned}$$

2. This question is also from pre-calculus. The equation below represents a circle, but it is not in the standard form of the equation of a circle. Complete the square in y to put this equation in the standard form for a circle.

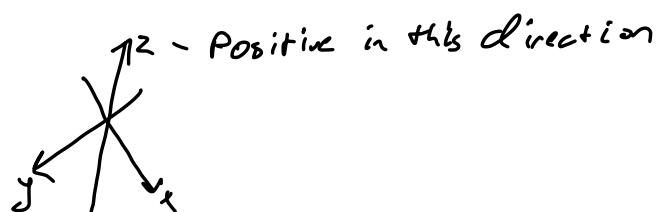
$$x^2 + y^2 - 6y = 16$$

$$\begin{aligned} \Rightarrow x^2 + y^2 - 6y + \left(\frac{6}{2}\right)^2 &= 16 + \left(\frac{6}{2}\right)^2 \\ \Rightarrow \boxed{x^2 + (y - 3)^2} &= 25 \end{aligned}$$

Definitions in Section 12.1: Give the definition of each term below, using the given passage from your text/e-text.

- **Right-Hand rule:** [beginning of section 12.1, before Figure 1.]

The positive z -axis goes up



- **Octants and First Octant:** [discussion surrounding figure 3 of section 12.1.]

The xy - xz - and yz -planes divide $\mathbb{R}^3$ into 8 octants, the 3D analogue of quadrants. The first octant is the set of points where x , y , and z are positive

- **Cartesian Product:** [non-boldfaced term following figure 6 of section 12.1.]

The cartesian product of two sets gives the ordered pairs that can be formed from those sets, that is $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$.

We generally use $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \mathbb{R}^3$ in this course, as $\mathbb{R}^3$ is the set of all ordered triples of real numbers.

- **Distance formula in three dimensions:** [formula above figure 13 of section 12.1.]

For $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$

$$\overline{P_1 P_2} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

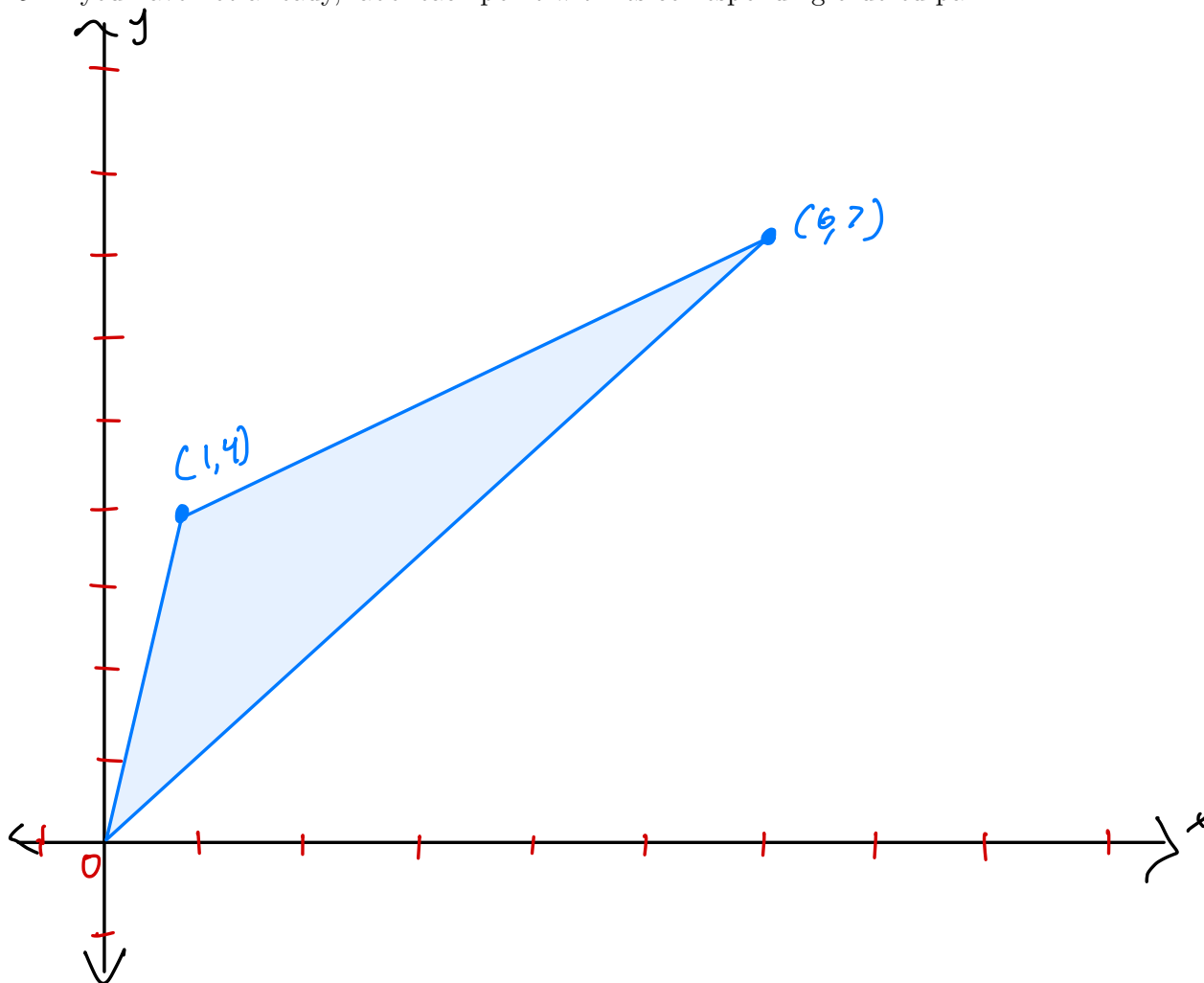
More generally, for $P, Q \in \mathbb{R}^n$, $\overline{PQ} = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$

12.2 Vectors

Prerequisite information for Section 12.2:

Sketch a two-dimensional xy coordinate axis in the space below, depicting x -values from -1 to 7 and y -values from -1 to 8. Complete each of the following steps, drawing on your graph. (We will use this diagram as part of the lecture during class.)

1. Plot the point $(1, 4)$ on your graph.
2. Draw a line segment from the origin $(0, 0)$ to the point $(1, 4)$.
3. Create a new point by “adding” point $(1, 4)$ to the point $(5, 3)$ as follows: We will define the **sum of these two points** to be the new point $(1 + 5, 4 + 3)$; we will call this **adding the points component-wise**.
4. Draw two more line segments. The first: from $(1, 4)$ to your new point $(6, 7)$. The second from the origin $(0, 0)$ to your new point $(6, 7)$. (Your picture should now show a triangle.)
5. If you have not already, label each point with its corresponding ordered pair.

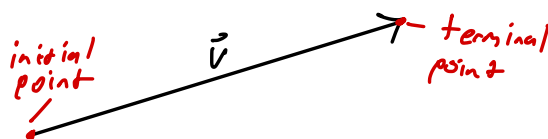


Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Vector:** [first paragraph of section 12.2.]

A mathematical quantity with both magnitude and direction.

- **Initial point and terminal point:** [discussion above figure 1 of section 12.2—it's ok to draw a picture]



- **Vector addition:** [box above figure 3 of section 12.2.]

If $\vec{u}$ and $\vec{v}$ are vectors positioned such that the terminal point of $\vec{u}$ is the initial point of $\vec{v}$, $\vec{u} + \vec{v}$ is the vector from the initial point of $\vec{u}$ to the terminal point of $\vec{v}$.

$$\vec{u} = \overrightarrow{PQ}, \vec{v} = \overrightarrow{QR} \Rightarrow \vec{u} + \vec{v} = \overrightarrow{PR}$$

- **Scalar multiplication:** [box above figure 7 of section 12.2.]

For $c \in \mathbb{R}$, $c\vec{v}$ is the vector whose length is $|c|$ times the length of $\vec{v}$, and whose direction is the same for $c > 0$ and opposite for $c < 0$. For $c = 0$ or $\vec{v} = \vec{0}$, $c\vec{v} = \vec{0}$.

- **Magnitude of a vector:** [discussion following example 3 of section 12.2.]

The magnitude of $\vec{v}$ is given by $|\vec{v}|$ or $\|\vec{v}\|$, and is calculated with the distance formula.

Topics to learn in this section:

1. The definition and length of vectors, vector addition, and scalar multiplication.
2. Unit vectors (and how to create one from any non-zero vector).
3. The relationship between the vector representation, $\vec{a} = \langle a_1, a_2 \rangle$, the point $P(a_1, a_2)$, and the position representation $\vec{OP}$.

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12.3 The Dot Product

Prerequisite information for Section 12.3:

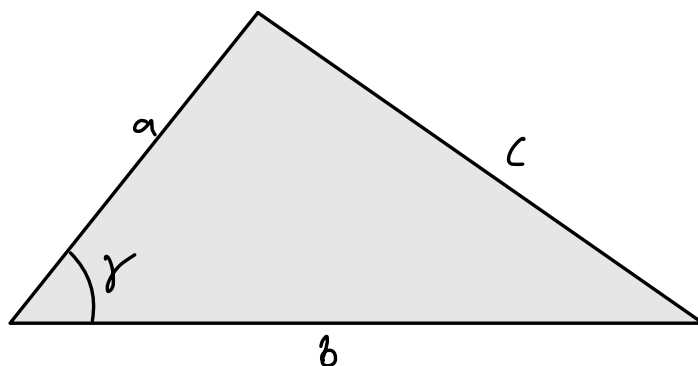
1. Complete the table below, giving **EXACT VALUES** (no calculator approximations).

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

2. State the **Law of Cosines**, drawing a labeled diagram of a triangle to illustrate.

according to google,

$$c = \sqrt{a^2 + b^2 - 2ab \cos \gamma}$$



Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Dot Product** (in 3 dimensions): [given in Box 1 of Section 12.3]

For $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$, we define

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

This is an example of an inner product on $\mathbb{R}^n$

We also have $\vec{a} \cdot \vec{b} = \|\vec{a}\| \|\vec{b}\| \cos \theta$

- **Orthogonal vectors:** [given after Example 3 of Section 12.3]

$\vec{a}$ and $\vec{b}$ are orthogonal if and only if

$$\vec{a} \cdot \vec{b} = 0$$

because this implies the angle they form is $\frac{\pi}{2}$

$$\therefore \cos \frac{\pi}{2} = 0$$

$\vec{0}$ is orthogonal to all vectors

- **Direction angles** (in 3 dimensions): [before Figure 3 of Section 12.3]

For some $\vec{a} = \langle a_1, a_2, a_3 \rangle$, we say the angles $\vec{a}$ forms with the axes are its direction angles α, β, γ .

$$\cos \alpha = \frac{a_1}{\|\vec{a}\|}, \cos \beta = \frac{a_2}{\|\vec{a}\|}, \cos \gamma = \frac{a_3}{\|\vec{a}\|}$$

Unit vector
↓

$$\frac{\vec{a}}{\|\vec{a}\|} = \langle \cos \alpha, \cos \beta, \cos \gamma \rangle \quad \text{and} \quad \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Topics to learn in this section:

1. The algebraic and geometric formulations of the dot product.
2. The interpretation of the sign of $\vec{a} \cdot \vec{b}$ in terms of the angle between $\vec{a}$ and $\vec{b}$.
3. Orthogonal vectors.
4. Vector and scalar projections.